# API documentation for Refund Integration with Enat Bank

Developer Guide

July 2025

# Refund API Documentation

# 1. Introduction

The Refund API provided by Ethiopian Airlines offers a standardized REST interface that enables third-party banks and financial institutions to interact with our ECOM service. The API supports seamless integration for refund processing, offering flexibility in both JSON and XML formats for requests and responses.

**Security**

Banks and PSP that access the API are subject to the security protections and user authentication that is provided by Ethiopian.

User Authentication is a combination of the followings:

- ✓ Basic Authentication encoded with Base64 in transit over HTTPS. Any Third party uses username: password provided specifically for this API. Please refer http://tools.ietf.org/html/rfc2617 for Basic Authentication.

# 2. Refund API Endpoints

## 2.1. Initiate Refund Request

In this process, Ethiopian Airlines initiates a refund request to the designated Enat bank endpoint (*the Bank shall configure and provide the endpoint to receive the refund initiation from ET*) using the following request parameters.

**Request Parameters:**

| Properties | Type | Required | Description |
|---|---|---|---|
| *Header* | *String* | *Yes* | *Basic authentication header* |
| | | | |
| shortCode | String | Yes | Specific unique Merchant code (for Ethiopian Airlines Ticket or other services) configured by FlyGate (ET) |
| OrderId | String | Yes | Original ET Order ID used for the payment |
| FirstName | String | Yes | Customer First Name |

| | | | |
|---|---|---|---|
| LastName | String | Yes | Customer Last Name |
| Amount | Double | Yes | The refundable amount |
| Currency | String | Yes | Currency in three-digit code (e.g., “ETB”) |
| **ReferenceNumber** | **String** | **Yes** | **Original Bank Payment reference number** |
| RefundFOP | String | Yes | Bank or Payment Option ID assigned by ET |
| **RefundReferenceCode** | **String** | **Yes** | **Unique refund reference code provided by FlyGate (ET)** |

**Sample Request**

```
{
  "shortCode": "526341",
  "orderId": "ABCDEF",
  "firstName": "Abebe",
  "lastName": "Kebede",
  “amount”: “100.00”,
  "currency": “ETB”,
  "ReferenceNumber": "FT1234566789"
 "refundFOP": "138",
  "refundReferenceCode": "EM24379446057FGHJHGFD"
}
```

### 2.1.1. Refund Initiate Response from the bank (Enat bank).

After initiating the refund request, the API will return the following attributes as acknowledgement response for accepting the requested refund:

| Attribute | Data Type | Required | Description |
|---|---|---|---|
| ResponseCode | Int | Yes | **1** for ***Success*** and **0** For ***Error*** |
| ResponseCodeDescription | string | Yes | Description of the response code |
| Status | string | Yes | Status of the refund process from the bank |

**Sample expected response for Success**

```
{
  "ResponseCode": 1,
  "ResponseCodeDescription": "Successfully accepted Refund request",
  "Status": "Success"
}
```

**Sample expected response for Failure**

```
{
    "ResponseCode": 0,
    "ResponseCodeDescription": "There is no any transaction associated with this {ReferenceNumber}",
    "Status": "Error"
}
```

## 2.2. Refund Confirmation Response

This API confirms the Refund of the request and updates the transaction status. Once the bank processes the refund, a confirmation request must be sent using this confirmation API.

**Endpoint:**

| METHOD | URL |
|---|---|
| POST | {{url}}/Enat/api/V1.0/Enat/ConfirmRefund |

**Request Parameter**

| Properties | Type | Required | Notes |
|---|---|---|---|
| *Header* | *String* | *Yes* | *Basic authentication header* |
| | | | |
| Shortcode | String | Yes | Specific unique Merchant code (for Ethiopian Airlines Ticket or other services) configured by FlyGate (ET) |
| orderId | String | Yes | Original ET Order ID used for payment |
| Amount | double | Yes | The refundable amount |
| Currency | String | Yes | Currency in three-digit code (e.g., “USD”) |
| **refundReferenceCode** | **String** | **Yes** | **Unique refund reference code from FlyGate** |
| **bankRefundReference** | **String** | **Yes** | **The new bank's reference number for the refunded transaction** |
| refundDate | String | Yes | Date of refund in YYYY-MM-DD format |
| refundAccountNumber | String | Yes | Customer's account number where the refund was sent |
| **AccountHolderName** | **String** | **Yes** | **Name of the customer holding the account** |
| RefundFOP | String | Yes | Unique Bank ID |
| Status | Int | Yes | 1 for *Success* and 0 For *Failure* |

| | | | |
|---|---|---|---|
| Remark | String | Yes | Any additional remarks or response descriptions |

**Sample Confirmation Request**


```
{
  "shortcode": "526341",
  "OrderId": "ABCDEF",
  "Amount": "100.00",
  "currency": "ETB",
  "RefundReferenceCode": "EM24379446057FGHJHGFD",
  "bankRefundReference": "FT12121212",
  "refundDate": "2024-02-14",
  "RefundAccountNumber": "013*******9500",
  "AccountHolderName": "Abebe Kebede"
  "refundFOP": "138",
  "Status": "1",
  "Remark": "Successfully Refunded",
}
```


### 2.2.1. Sample FlyGate Response (From ET)

**Successful Refund Confirmation:**

Confirm refund response for **success** refund confirmation.


```
{
  "refundReferenceCode": "EM24379446057FGHJHGFD ",
  "bankRefundReference": "FT12121212",
  "OrderId": "ABCDEF",
  "Amount": "100.00",
  "status": 1,
  "message": "Operation Success"
}
```


**Error in Refund Confirmation:**

Confirm refund response if **Error** happened.


```
{
  "refundReferenceCode": "EM24379446057FGHJHGFD",
  "bankRefundReference": "FT12121212",
  "OrderId": "ABCDE",
  "Amount": "100.00",
  "status": 0,
  "message": "Order Already Refunded"
```

}

# 3. Environments

The API is available in both CERT and PROD environments:

- **CERT Environment (Testing) :**
  - VPN URL: **https://flytest.ethiopianairlines.com**
  - Public URL: **https://ethiopiangatewaytest.azurewebsites.net/**
- **PROD Environment (Production)**:
  - VPN URL: *to be advised soon*

# 4. Short Code

For integration, communication, merchant identification, and account mapping, each message within API requests and responses include Short Code. These identifiers enable streamlined technical communication and precise mapping of transactions.

The tables below outline the designated Short Codes for both Test and Production environments. On this integration the below short codes are specific to Ethiopian Airlines Passenger Ticketing service.

### 4.1. Test Environment: Short Code

| No | Merchant Name | Short Code |
|---|---|---|
| 1 | Ethiopian Airlines Passenger Ticketing | **526341** |

### 4.2. Production Environment: Short Code

| No | Merchant Name | Short Code |
|---|---|---|
| 1 | Ethiopian Airlines Passenger Ticketing | **858682** |

# 5. Additional Information

- Ensure all requests are made over secure **HTTPS** to maintain data security.

- Always validate and sanitize inputs before making requests to avoid potential issues.